

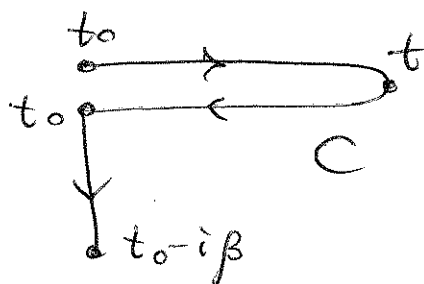
Lecture 5

Time-ordered integrals

1. Here we should learn how to replace $\int_C$ by real-time (ordinary) integrals. We first consider

$$c(t, t') = \int_C a(t, \tau) b(\tau, t') d\tau \quad (1)$$

where a, b, c are objects of two times defined on the contour



Generally, these objects may have 3 parts; e.g.

$$a(t, t') = \delta_{tt'} a^\delta(t) + \theta_{tt'} a^>(t, t') + \theta_{t't} a^<(t, t') \quad (2)$$

Our task is to substitute (2) for a and b into (1) and find three components of $c(t, t')$:

$$\begin{aligned} C &= \int_C \underbrace{\left[\delta_{t\tau} a^\delta_t + \theta_{t\tau} a^>_{t\tau} + \theta_{\tau t} a^<_{\tau t} \right]}_{a_{t\tau}} \underbrace{\left[\delta_{\tau t'} b^\delta_{\tau t'} + \theta_{\tau t'} b^>_{\tau t'} + \theta_{t'\tau} b^<_{t'\tau} \right]}_{b_{\tau t'}} d\tau \\ &\Rightarrow a_t^\delta b_{tt'} + \cancel{a_t^\delta b_{tt'}} + \int_C \left\{ \theta_{t\tau} \theta_{\tau t'} a^>_{t\tau} b^>_{\tau t'} \right. \\ &\quad + \theta_{t\tau} \theta_{t'\tau} a^>_{t\tau} b^<_{\tau t'} + \theta_{\tau t} \theta_{\tau t'} a^<_{\tau t} b^>_{\tau t'} \\ &\quad \left. + \theta_{\tau t} \theta_{t'\tau} a^<_{\tau t} b^<_{\tau t'} \right\} d\tau \end{aligned} \quad (3)$$

• $t > t'$ and are both on the real part of the contour.

$$C(t, t') \rightarrow C^>(t, t') = a^> b^> + a^> b^< + c^>> + c^>< + c^<> + c^<< \quad (4)$$

where:

$$C^>>(t, t') = \int_C d\tau \underbrace{\theta_{t\tau}}_{\tau < t} \underbrace{\theta_{\tau t'}}_{\tau > t'} a^>_{t\tau} b^>_{\tau t'} = \int_{t'}^t d\tau a^>_{t\tau} b^>_{\tau t'} = \int_{t_0}^t - \int_{t_0}^{t'} \quad (5)$$

$$C^><(t, t') = \int_C d\tau \underbrace{\theta_{t\tau}}_{\tau < t} \underbrace{\theta_{\tau t'}}_{\tau < t'} a^>_{t\tau} b^<_{\tau t'} = \int_{t_0}^{t'} d\tau a^>_{t\tau} b^<_{\tau t'} \quad (6)$$

$$C^<>(t, t') = \int_C d\tau \underbrace{\theta_{t\tau}}_{\tau > t} \underbrace{\theta_{\tau t'}}_{\tau > t'} a^<_{t\tau} b^>_{\tau t'} = \int_t^{t_0-i\beta} d\tau a^<_{t\tau} b^>_{\tau t'} = \quad (7)$$

$$= \int_t^{t_0} d\tau a^<_{t\tau} b^>_{\tau t'} + \int_{t_0}^{t_0-i\beta} d\tau a^<_{t\tau} b^>_{\tau t'} \quad (7)$$

$$C^<<(t, t') = \int_C d\tau \underbrace{\theta_{t\tau}}_{\tau > t} \underbrace{\theta_{\tau t'}}_{\tau < t'} a^<_{t\tau} b^<_{\tau t'} = 0$$

Here:

$$a^\Gamma(t, \tau) \equiv a^<(t, t_0 - i\tau), \quad b^\Gamma(\tau, t) = b^>(t_0 - i\tau, t) \quad (8)$$

$$\begin{aligned} C^>> + C^>< + C^<> + C^<< &= \left[\int_{t_0}^t d\tau a^>_{t\tau} b^>_{\tau t'} - \int_{t_0}^{t'} d\tau a^>_{t\tau} b^>_{\tau t'} \right] \\ &+ \int_{t_0}^{t'} d\tau a^>_{t\tau} b^<_{\tau t'} + \int_{t_0}^t d\tau a^<_{t\tau} b^>_{\tau t'} + \int_{t_0}^{t_0-i\beta} d\tau a^\Gamma_{t\tau} b^\Gamma_{\tau t'} \\ &= \int_{t_0}^t (a^>_{t\tau} - a^<_{t\tau}) b^>_{\tau t'} + \int_{t_0}^{t'} a^>_{t\tau} (b^>_{\tau t'} - b^<_{\tau t'}) \end{aligned}$$

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$$+ \int_{t_0}^{t_0 - i\beta} dt' a_{t\tau}^T b_{\tau t'}^\Gamma \quad (9)$$

since $t > \tau$

Here $a_{t\tau}^> - a_{t\tau}^< = \theta_{t\tau} (a_{t\tau}^> - a_{t\tau}^<) =$ } line in $a_{t\tau}^R$
 $b_{\tau t'}^> - b_{\tau t'}^< = \theta_{t\tau} (b_{\tau t'}^> - b_{\tau t'}^<) \quad \left\{ \text{and } b_{\tau t'}^A \right.$

We introduce complete definitions:

$$\begin{aligned} a^R(t, t') &= a^\delta(t) \delta_{tt'} + \theta_{tt'} (a^>(t, t') - a^<(t, t')) \\ a^A(t, t') &= a^\delta(t) \delta_{tt'} - \theta_{t't} (a^>(t, t') - a^<(t, t')) \end{aligned} \quad (10)$$

Then:

$$\begin{aligned} C_{tt'}^> &= \cancel{a_{tt'}^> b_{tt'}^\delta} + \cancel{a_{tt'}^\delta b_{tt'}^>} + \int_{t_0}^t d\tau [a_{t\tau}^R - a_{t\tau}^\delta \delta_{t\tau}] b_{\tau t'}^> + \\ &+ \int_{t_0}^{t'} d\tau a_{t\tau}^> [b_{\tau t'}^A - \cancel{b_{\tau t'}^\delta \delta_{\tau t'}}] + \int_{t_0}^{t_0 - i\beta} d\tau a_{t\tau}^T b_{\tau t'}^\Gamma \\ &= \underbrace{\int_{t_0}^t d\tau a_{t\tau}^R b_{\tau t'}^>}_{t > \tau \rightarrow \text{upper boundary } t_0 + \infty} + \underbrace{\int_{t_0}^{t'} d\tau a_{t\tau}^> b_{\tau t'}^A}_{\tau < t' \rightarrow \text{upper boundary } t_0 + \infty} + \int_{t_0}^{t_0 - i\beta} d\tau a_{t\tau}^T b_{\tau t'}^\Gamma \end{aligned}$$

$\hookrightarrow$

$$C^>(t, t') = \int_{t_0}^{\infty} d\tau [a^R(t, \tau) b^>(\tau, t') + a^>(t, \tau) b^A(\tau, t')] + \int_{t_0}^{t_0 - i\beta} d\tau' a^T(t, \tau') b^\Gamma(\tau', t') \quad (11)$$

Symbolically: $C^> = a^R \cdot b^> + a^> \cdot b^A + a^T \star b^\Gamma \quad (12)$

The last term in more detail:

$$\int_{t_0}^{t_0-i\beta} d\tau' a^\dagger(t, \tau') b^\dagger(\tau', t') = \left| \begin{array}{l} t' = t_0 - i\tau \\ d\tau' = -i d\tau \end{array} \right|$$

$$= -i \int_0^\beta d\tau a^\dagger(t, t_0 - i\tau) b^\dagger(t_0 - i\tau, t') = -i \int_0^\beta d\tau a^\dagger(t, \tau) b^\dagger(\tau, t') \quad (13)$$

• $t' < t$ and both on the real part. $\triangleright$

$$\boxed{C^< = a^R \cdot b^< + a^< \cdot b^A + a^\dagger \star b^\dagger} \quad (14)$$

• For other times we get similarly:

$$\boxed{C^\dagger(\tau, t') = a^\dagger \cdot b^A + a^M \star b^\dagger} \quad (15)$$

where

$$\boxed{a^M(\tau, \tau') = i a^\delta(\tau) \delta_{\tau\tau'} + \theta_{\tau\tau'} \underbrace{a^\dagger(\tau, \tau')}_{a^\dagger(t_0-i\tau, t_0-i\tau')} + \theta_{\tau'\tau} \underbrace{a^<(\tau, \tau')}_{a^<(t_0-i\tau, t_0-i\tau')}} \quad (16)$$

is the Matsubara's function.

$$\bullet \boxed{C^\dagger(t, \tau) = a^R \cdot b^\dagger + a^\dagger \star b^M} \quad (17)$$

$$\boxed{C^M = a^M \star b^M} \quad (18)$$

$$\boxed{C^R = a^R \cdot b^R} \quad \boxed{C^A = a^A \cdot b^A}$$

• Products of 3 and more functions can be considered similarly:

$$D(1, 2) = \int_c d3 \int_c d4 A(1, 3) B(3, 4) C(4, 2) \rightarrow$$

results in

$$D^K = A^R \cdot (B^R \cdot C^K + B^K \cdot C^A) + A^K \cdot B^A \cdot C^A + A^R \cdot B^T \star C^T + A^T \star (B^T \cdot C^A + B^A \star C^T) \quad (19)$$

Others follow similarly, ~~eq.~~

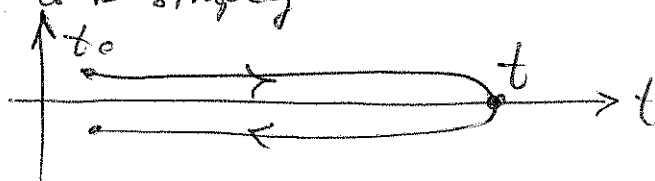
2. Keldysh equations

Start from the Dyson eq: $G = G_0 + G_0 \Sigma G$

~~$G^K = G_0^K + G_0^K \Sigma^K G^K$~~ and ignore the imaginary part of the contour. This corresponds to using $e^{-\beta H_0}$ in all averaging, i.e.

$$\langle T_C \psi_{ye}(1) \psi_{ye}^\dagger(2) \rangle = \frac{\text{Tr}[e^{-\beta H_0} T_C \psi_{ye}(1) \psi_{ye}^\dagger(2)]}{\text{Tr}[e^{-\beta H_0}]} \quad (20)$$

and the contour is simply



Then,

$$G^K = G_0^K + G_0^R \cdot \Sigma^R G^K + G_0^R \Sigma^K G^A + G_0^K \Sigma^A G^A \quad (21)$$

~~$$(G_0^K + G_0^R \Sigma^R G^K + G_0^R \Sigma^K G^A + G_0^K \Sigma^A G^A) + (G_0^R \Sigma^K) G^K \quad (22)$$~~

~~Iterate it with respect to G^K :~~

~~$$(1 - G_0^R \Sigma^R) G^K = G_0^K (1 + \Sigma^A G^A) + G_0^R \Sigma^K G^A$$~~

$$(1 - G_0^R \Sigma^R) G^K = G_0^K (1 + \Sigma^A G^A) + G_0^R \Sigma^K G^A \quad (22)$$

Next, write the Dyson eq. for R and A: For these objects

$$(ABC)^R = A^R \cdot B^R \cdot C^R \quad \left(\begin{array}{l} \text{excluding} \\ \text{the complex part of } \Sigma \end{array} \right)$$

and the same for $(ABC)^A$. Therefore,

$$G^R = G_0^R + G_0^R \Sigma^R G^R \quad \text{or} \quad = G_0^R + G^R \Sigma^R G_0^R$$

$$\hookrightarrow G^R(1 - \Sigma^R G_0^R) = G_0^R \quad \text{or} \quad (1 - G_0^R \Sigma^R) G^R = G_0^R$$

and hence

$$(1 - G_0^R \Sigma^R)^{-1} = G_0^R (G^R)^{-1}$$

$$\hookrightarrow (1 - G_0^R \Sigma^R)^{-1} = G^R (G_0^R)^{-1} = (G_0^R + G_0^R \Sigma^R G_0^R) (G_0^R)^{-1} \\ = 1 + G^R \Sigma^R$$

Multiply (22) from the left by $(1 - G_0^R \Sigma^R)^{-1}$:

$$\boxed{G^< = (1 + G^R \Sigma^R) G_0^< (1 + \Sigma^A G^A) + G_0^R \Sigma^< G^A} \quad (23)$$

Also, similarly,

$$\boxed{G^> = (1 + G^R \Sigma^R) G_0^> (1 + \Sigma^A G^A) + G^R \Sigma^> G^A} \quad (24)$$

3. Steady-state regime (Ψ does not depend on t)

$\hookrightarrow$ GFs depend only on the time difference. if you wait sufficiently long (the processes no longer depend on time, but they are still out of equilibrium).

We show that the 1st term in (24) is zero in this case. But first we need to work out all GF's we need in the ω -representation (FT). $G_0 \rightarrow$ is based on a one-part. Hamiltonian $\hat{H}$

$$G_0^>(1,2) \rightarrow G_0^>(x_1 t_1, x_2 t_2) = -i \sum_j \chi_j(x_1) \chi_j^*(x_2) (1 - f_j) e^{-i\bar{\epsilon}_j(t_1 - t_2)}$$

$$G_0^>(x_1, x_2, \omega) = \int dt_1 dt_2 G_0^>(x_1 t_1, x_2 t_2) e^{i\omega(t_1 - t_2)}$$

$$= -i \sum_j \chi_j(x_1) \chi_j^*(x_2) (1 - f_j) 2\pi \delta(\omega - \bar{\epsilon}_j)$$

$$= -2\pi i f(\omega) \left[\sum_j \chi_j(x_1) \chi_j^*(x_2) \delta(\omega - \bar{\epsilon}_j) \right] \quad (25)$$

$$G_0^<(x_1, x_2, \omega) = \int d(t_1 - t_2) \left[i \sum_j \chi_j(x_1) \chi_j^*(x_2) f_j e^{-i\bar{\epsilon}_j(t_1 - t_2)} \right] e^{i\omega(t_1 - t_2)}$$

$$= 2\pi i f(\omega) \left[\sum_j \chi_j(x_1) \chi_j^*(x_2) \delta(\omega - \bar{\epsilon}_j) \right] \quad (26)$$

$$G_0^A(x_1, x_2, \omega) = \int d(t_1 - t_2) \left[i \theta(t_2 - t_1) \sum_j \chi_j(x_1) \chi_j^*(x_2) e^{-i\bar{\epsilon}_j(t_1 - t_2)} \right] e^{i\omega(t_1 - t_2)}$$

$$= i \sum_j \chi_j(x_1) \chi_j^*(x_2) \int_{-\infty}^{\infty} d\tau \theta(\tau) e^{i(\omega - \bar{\epsilon}_j)\tau}$$

Here

$$\int d\tau \rightarrow \int_{-\infty}^0 e^{i(\omega - \bar{\epsilon}_j)\tau} d\tau \rightarrow \lim_{\delta \rightarrow +0} \int_{-\infty}^0 e^{i(\omega - i\delta - \bar{\epsilon}_j)\tau} d\tau$$

ensures convergence at $\tau \rightarrow -\infty$

$$= \lim_{\delta \rightarrow +0} \frac{e^{i(\omega - \bar{\epsilon}_j)\tau}}{i(\omega - \bar{\epsilon}_j)} \Big|_{-\infty}^0 = \lim_{\delta \rightarrow +0} \frac{e^{i(\omega - i\delta - \bar{\epsilon}_j)\tau}}{i(\omega - i\delta - \bar{\epsilon}_j)} \Big|_{-\infty}^0$$

$$= \lim_{\delta \rightarrow +0} \frac{1}{i(\omega - \bar{\epsilon}_j)} \text{ or simply } \frac{1}{i(\omega - \bar{\epsilon}_j)} \quad (27)$$

$$\hookrightarrow G_0^A(x_1, x_2, \omega) = \sum_j \frac{\chi_j(x_1) \chi_j^*(x_2)}{\omega - \bar{\epsilon}_j}, \quad \omega^- = \omega - i\delta \quad (28)$$

Similarly

$$G_0^R(x_1, x_2, \omega) = + \sum_j \frac{\chi_j(x_1) \chi_j^*(x_2)}{\omega^+ - \bar{\epsilon}_j}, \quad \omega^+ = \omega + i\delta \quad (29)$$

We also have to calculate the inverse GF defined as:

$$\int G^{-1}(1,2) G(2,3) d2 = \delta(1,3)$$

Take the FT:

$$\int G^{-1}(x_1, x_2, \omega) G(x_2, x_3, \omega) dx_2 = \delta(x_1 - x_3) \quad (30)$$

Then, it is easy to see, that if, say,

$$A(x_1, x_2, \omega) = \sum_j \chi_j(x_1) \chi_j^*(x_2) g_j(\omega).$$

then

$$A^{-1}(x_1, x_2, \omega) = \sum_j \chi_j(x_1) \chi_j^*(x_2) g_j^{-1}(\omega)$$

Indeed,

$$\int A^{-1}(x_1, x_2, \omega) A(x_2, x_3, \omega) dx_2 =$$

$$= \sum_{j, j'} \chi_j(x_1) \chi_{j'}^*(x_3) g_{j'}(\omega) g_{j'}^{-1}(\omega) \underbrace{\int \chi_{j'}^*(x_2) \chi_{j-1}(x_2) dx_2}_{\delta_{jj'}}$$

$$= \sum_j g_j(\omega) g_j^{-1}(\omega) \chi_j(x_1) \chi_j^*(x_3) \stackrel{\text{should be}}{=} \delta(x_1 - x_3) = \sum_j \chi_j(x_1) \chi_j^*(x_3)$$

which will only happen if $g_j^{-1}(\omega) = \frac{1}{g_j(\omega)}$.

Therefore,

$$G_o^R(x_1, x_2, \omega)^{-1} = \sum_j (\omega^+ - \bar{\epsilon}_j) \chi_j(x_1) \chi_j^*(x_2) \quad (31)$$

$$G_o^A(x_1, x_2, \omega)^{-1} = \sum_j (\omega^- - \bar{\epsilon}_j) \chi_j(x_1) \chi_j^*(x_2) \quad (32)$$

With these results, we can now rearrange the 1st term in (24):

$$G^R = G_o^R + G_o^R \Sigma^R G_o^R \rightarrow 1 + G^R \Sigma^R = G^R (G_o^R)^{-1} \rightarrow$$

also,

$$G^A = G_o^A + G_o^A \Sigma^A G_o^A \rightarrow 1 + \Sigma^A G^A = (G_o^A)^{-1} G^A$$

↳

$$(1 + G^R \Sigma^R) G_o^R (1 + \Sigma^A G^A) = G^R \left[(G_o^R)^{-1} G_o^R (G_o^A)^{-1} \right] G^A \quad (33)$$

Consider [...] above:

$$[...] = (G_o^R)^{-1} G_o^> (G_o^A)^{-1}$$

In a steady-state an integration of products is a convolution, which in the ω -representation becomes a product of functions:

$$\begin{aligned} [...]_{\omega} &= \cancel{\sum_j (\omega^+ - \bar{\epsilon}_j)} \int dx_2 dx_3 \left[\sum_j (\omega^+ - \bar{\epsilon}_j) \chi_j(x_1) \chi_j^*(x_2) \right] \times \\ &\times \left[2\pi i \cancel{f(\omega)} \sum_{j'} \chi_{j'}(x_2) \chi_{j'}^*(x_3) \delta(\omega - \bar{\epsilon}_{j'}) \right] \times \left[\sum_{j''} (\omega - \bar{\epsilon}_{j''}) \chi_{j''}(x_3) \chi_{j''}^*(x_4) \right] \\ &= \left| \begin{array}{l} \delta_{jj'} \text{ due to } \int dx_2 \\ \delta_{jj''} \text{ due to } \int dx_3 \end{array} \right| = \sum_j (\omega^+ - \bar{\epsilon}_j) 2\pi f(\omega) (\omega - \bar{\epsilon}_j) \delta(\omega - \bar{\epsilon}_j) \chi_j(x_1) \chi_j^*(x_4) \end{aligned}$$

But

$$\begin{aligned} (\omega^+ - \bar{\epsilon}_j)(\omega - \bar{\epsilon}_j) \delta(\omega - \bar{\epsilon}_j) &= (\omega + i\delta - \bar{\epsilon}_j)(\omega - i\delta - \bar{\epsilon}_j) \delta(\omega - \bar{\epsilon}_j) \\ &= [(\omega - \bar{\epsilon}_j)^2 + \delta^2] \delta(\omega - \bar{\epsilon}_j) \rightarrow \delta^2 \cdot \delta(\omega - \bar{\epsilon}_j) \rightarrow 0 \text{ as } \delta \rightarrow 0 \end{aligned}$$

Therefore, for steady-state processes:

$$\boxed{G^> = G^R \Sigma^> G^A, \quad G^< = G^R \Sigma^< G^A} \quad (34)$$

4. Kadanoff-Baym equation

Using $\Sigma(1,2) = \underbrace{\Sigma_{HF}(1,2)}_{\sim \delta(t_1-t_2)} + \theta(t_1, t_2) \Sigma^>(1,2) + \theta(t_2, t_1) \Sigma^<(1,2) \quad (35)$

one can write down eqs. for each component of G^i :

$$(i\partial_{t_1} - h_1) G(1,2) = \delta(1,2) + \int_C \Sigma(1,3) G(3,2) d3$$

where $C(1,2) = \int_C \Sigma(1,3) G(3,2) d3$

is decomposed for each component of C (and hence G)

using Langreth rules:

$$\begin{cases} C^>(1,2) = \Sigma^R \cdot G^> + \Sigma^? \cdot G^A + \Sigma^I \star G^I \\ C^< = \Sigma^R \cdot G^< + \Sigma^< \cdot G^A + \Sigma^I \star G^I \\ C^I = \Sigma^R \cdot G^I + \Sigma^I \star G^M \\ C^I = \Sigma^I \cdot G^A + \Sigma^M \star G^M \\ C^M = \Sigma^M \star G^M \end{cases} \quad \begin{matrix} \text{(G - does not have the} \\ \text{equal time component)} \end{matrix} \quad (36)$$

resulting in:

$$(i\partial_{t_1} - h_1) G^<(1,2) = \Sigma^R \cdot G^< + \Sigma^< \cdot G^A + \Sigma^I \star G^I \equiv C^<(1,2)$$

(note that $\delta(1,2)$ can be dropped as here $t_1 < t_2$ (strictly))

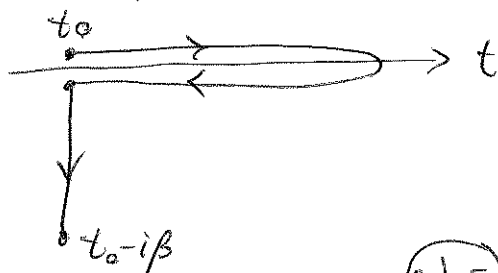
$$(i\partial_{t_1} - h_1) G^>(1,2) \equiv C^>(1,2) \quad (37)$$

$$(i\partial_{t_1} - h_1) G^I(1,2) = C^I(1,2)$$

$$(i\partial_{t_1} - h_1) G^M(1,2) = \delta(1,2) + C^M(1,2)$$

It is assumed here that $\Sigma^<,>$ can be expressed via G to close the set of eqs.

5. Idea of how the KB equations are solved.



After the $\int_C$ is replaced by actual real time integration, then there is no distinction anymore of whether it is on the upper or lower contour, etc.

Therefore:

$$\begin{cases} G^>(t_0, t_0) \equiv G^M(t_0 - i0^+, t_0 - i0) \equiv G^M(0^+, 0) \\ G^<(t_0, t_0) \equiv G^M(0, 0^+) \end{cases} \quad (38)$$

Also

$$\begin{cases} G^T(\tau, t_0) \equiv G^M(\tau, 0) \\ G^T(t_0, \tau) \equiv G^M(0, \tau) \end{cases} \quad (39)$$

and there are also the boundary conditions:

$$\begin{cases} G(x_1, t_0; x_2, t_2) \equiv -G(x_1, t_0 - i\beta; x_2, t_2) \\ G(x_1, t_1; x_2, t_0) \equiv -G(x_1, t_1; x_2, t_0 - i\beta) \end{cases} \quad (40)$$

In addition, there are some other identities:

$$G^<(1, 2)^* = -G^<(2, 1) \quad G^>(1, 2)^* = -G^>(2, 1) \quad (41)$$

and also equal-times one, e.g.:

$$\begin{aligned} G^>(x_1, t, x_2, t) &= -\frac{i}{Z} \langle \psi_{\mathcal{H}}(x_1, t) \psi_{\mathcal{H}}^{\dagger}(x_2, t) \rangle \\ &= -\frac{i}{Z} \text{Tr} \left\{ e^{-\beta \mathcal{H}_0} u(t_0, t) \psi(x_1) \underbrace{u(t, t)}_{=1} \psi^{\dagger}(x_2) u(t, t_0) \right\} \\ &= -\frac{i}{Z} \sum_{\lambda} e^{-\beta E_{\lambda}^0} \langle \lambda | u(t_0, t) \psi_{x_1} \psi_{x_2}^{\dagger} u(t, t_0) | \lambda \rangle \end{aligned}$$

Also,

$$G^<(x_1, t, x_2, t) \equiv +\frac{i}{Z} \sum_{\lambda} e^{-\beta E_{\lambda}^0} \langle \lambda | u(t_0, t) \psi_{x_2}^{\dagger} \psi_{x_1} u(t, t_0) | \lambda \rangle$$

with

$$Z = \sum_{\lambda} e^{-\beta E_{\lambda}^0}$$

Then,

$$\begin{aligned} G^> - G^< &= -\frac{i}{Z} \sum_{\lambda} e^{-\beta E_{\lambda}^0} \langle \lambda | u(t_0, t) \overbrace{(\psi_{x_1} \psi_{x_2}^{\dagger} + \psi_{x_2}^{\dagger} \psi_{x_1})}^{\delta(x_1 - x_2)} u(t, t_0) | \lambda \rangle \\ &= -\frac{i}{Z} \delta(x_1 - x_2) \underbrace{\sum_{\lambda} e^{-\beta E_{\lambda}^0} \langle \lambda | u(t_0, t) u(t, t_0) | \lambda \rangle}_1 \equiv -i\delta(x_1 - x_2) \end{aligned}$$

$$\hookrightarrow \boxed{G^>(x_1, t, x_2, t) = -i\delta(x_1 - x_2) + G^<(x_1, t, x_2, t)} \quad (42)$$

Time-stepping procedure:

- $G^M(t_1, t_2)$ - calculated first ("initial correlation") $\rightarrow$ standard equilibrium techniques can be used

All GFs are expanded in a basis, e.g.

$$G^M(t_1, t_2) = \sum_{ij} \chi_i(x_1) \chi_j^*(x_2) \underbrace{G_{ij}^M(t_1, t_2)}_{\text{only depends on "times"}}$$

$\hookrightarrow$ into the Σ and the KB equation; we obtain time-only DEs for various GF's. We assume some relationship between Σ and G ; basically, Σ is some functional of G .

- $G^<(1,2)^* = -G^<(2,1) \rightarrow$ it is sufficient to consider $t_2 > t_1$ in calculating $G^<(1,2)$

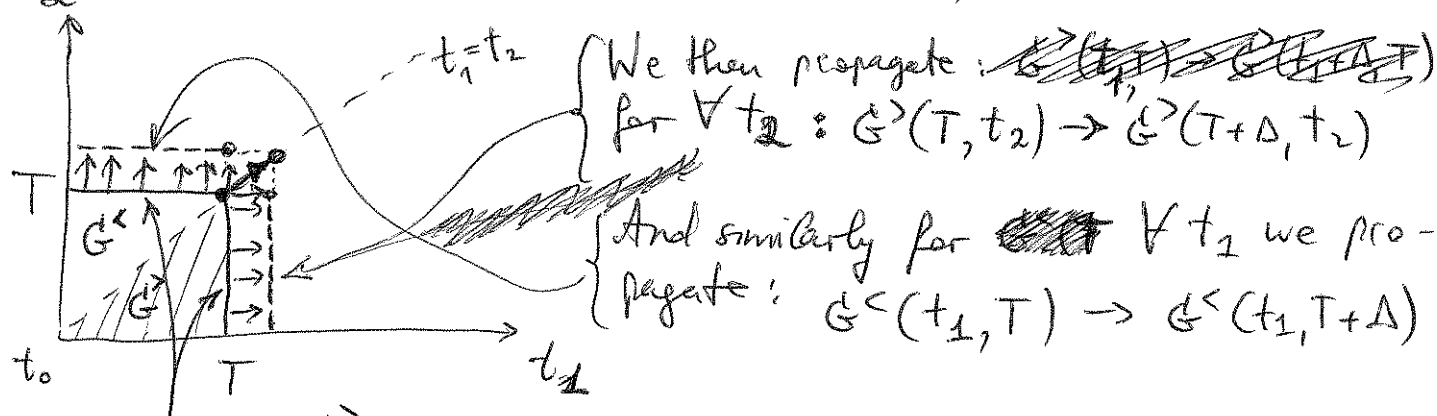
$$G^>(1,2)^* = -G^>(2,1) \rightarrow \text{only } G^>(1,2) \text{ for } t_1 > t_2 \text{ is enough}$$

- From EoM

$$(i\partial_{t_1} - h_1) G^>(1,2) = C^>(1,2) \text{ and } t_1 > t_2$$

we can time-step here in the 1st time argument:

$$G^>(t_1, t_2) \rightarrow G^>(t_1 + \Delta, t_2), \text{ small } \Delta$$



we know $G^>$ at the solid lines $t_1 = t_2 = T$

Also, a diagonal propagation

$$G^<(T, T) \rightarrow G^<(T + \Delta, T + \Delta)$$

$$G^>(T, T) \rightarrow G^>(T + \Delta, T + \Delta)$$

are required. This is done from DE for $i\partial_t G^>(x_1, t, x_2, t)$, etc.

The required "diagonal" propagation can be obtained from the KB equations for different times since

$$i\partial_t G^>(x_1 t, x_2 t) = \underbrace{\left[i\partial_{t_1} G^>(x_1 t_1, x_2 t_2) \right]}_{\text{KB eq. with differentiating wrt 1st argument}} + \underbrace{i\partial_{t_2} G^>(x_1 t_1, x_2 t_2)}_{\text{wrt to the 2nd argument}} \Big|_{t_1=t_2=t}$$

$$i\partial_{t_1} G^>(1,2) = h(1) G^>(1,2) + C_1^>(1,2)$$

$$-i\partial_{t_2} G^>(1,2) = h_2 G^>(1,2) + \boxed{C_2^>(1,2)} \quad \leftarrow \begin{array}{l} \text{obtained similarly} \\ \text{to the 1st eq. using} \\ \text{length rules} \end{array}$$

Since

$$-i\partial_{t_2} G(1,2) = h(2) G(1,2) = \delta_{12} + \int d3 \ G_{13} \varepsilon_{31}$$

Therefore,

$$\begin{aligned} i\partial_t G^>(x_1 t, x_2 t) &= \left[h_1 G^>(1,2) + C_1^>(1,2) \right]_{t_2=t_1=t} - \left[h_2 G^>(1,2) + C_2^>(1,2) \right]_{t_1=t_2=t} \\ &= (h_1 - h_2) G^>(1,2) \Big|_{t_2=t_1=t} + \left[C_1^>(1,2) - C_2^>(1,2) \right]_{t_1=t_2=t} \end{aligned} \quad (43)$$